

Unit IV

CLASSIFICATION OF COMPRESSOR

Air compressors may be classified as follows:

- 1) *According to design and principle of operation*
 - a) Reciprocating compressors
 - b) Rotary compressors
- 2) *According to action*
 - a) Single acting compressors
 - b) Double acting compressors
- 3) *According to number of stages*
 - a) Single stage compressors
 - b) Multistage compressors
- 4) *According to pressure limit*
 - a) Low pressure compressor
 - b) Medium pressure compressors
 - c) High pressure compressors
- 5) *According to capacity*
 - a) Low capacity compressors (Volume delivered $0.12\text{m}^3/\text{s}$ or less)
 - b) Medium capacity compressors (volume delivered $0.15\text{m}^3/\text{s}$ to $5\text{m}^3/\text{s}$)
 - c) High capacity compressors (Volume delivered is above $5\text{m}^3/\text{s}$)

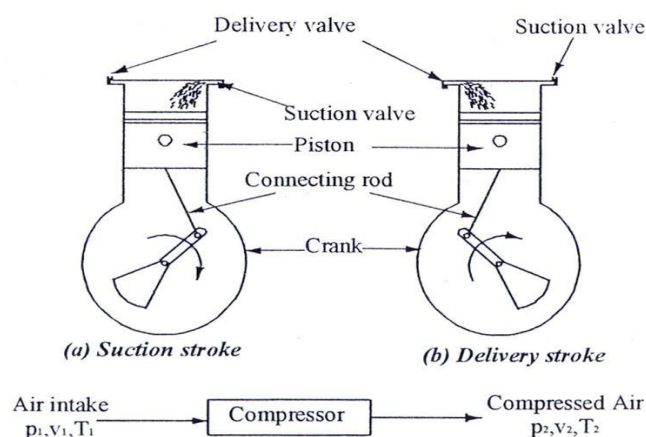
Single stage compressor:

In single stage compressor, the compression of the air from the initial pressure to the final pressure is carried out in one cylinder only.

Multistage compressor:

In multistage compressor, the compression of the air from the initial pressure to the final pressure is carried out in more than one cylinder.

WORKING OF SINGLE STAGE RECIPROCATING AIR COMPRESSOR



Single stage reciprocating air compressor

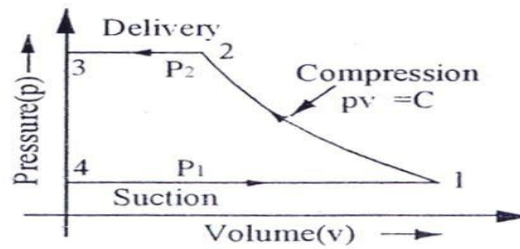
In a single stage compressor, the compression of air from the initial pressure to final pressure is carried out in one cylinder only. A schematic diagram of single stage, single acting compressor is shown in fig

It consists of a cylinder, piston, connecting rod, crank, inlet and discharge valves. When the piston moves downward i.e. during suction stroke, the pressure of air inside the cylinder falls below the atmospheric pressure. So the inlet valve opens and the air from atmospheric is sucked into the cylinder until the piston reaches the bottom dead center. During this stroke delivery valve remains closed. When the piston moves upwards both valves are closed. So the pressure inside the cylinder goes on increasing till it reaches required discharge pressure. At this stage, the discharge valve opens and the compressed air is delivered through this valve. Thus the cycle is repeated.

The work done by a single stage reciprocating air compressor without clearance volume

a) **Work done isothermal compression ($pv = c$):**

The p-v diagram for a single stage single acting reciprocating air compressor is shown in fig. the sequences of operation as represented on the diagram are as follows:



Isothermal compression

Process 4-1 => Represents the suction of air at pressure p_1

Process 1-2 => Air is compressed isothermally from pressure p_1 to pressure p_2

Process 2-3 => Represents the discharge of air at pressure p_2

Work done = Area 1 – 2 – 3 – 4 – 1

$$W = W_{comp} + W_{Delivery} - W_{Suction}$$

$$= p_1 v_1 \ln \left[\frac{v_1}{v_2} \right] + p_2 v_2 - p_1 v_1$$

$$[\text{For Isothermal process } W_{(comp)} = p_1 v_1 \ln \left[\frac{v_1}{v_2} \right]]$$

$$= p_1 v_1 \ln \left[\frac{v_1}{v_2} \right] + p_2 v_2 - p_2 v_2$$

$$[\text{For Isothermal process } p_1 v_2 - p_1 v_1]$$

$$W = p_1 v_1 \ln \left[\frac{v_1}{v_2} \right]$$

We know that,

$$p_1 v_1 = p_2 v_2$$

$$\frac{v_1}{v_2} = \frac{p_2}{p_1}$$

Substituting in equation (1)

$$W = p_1 v_1 \ln \left[\frac{p_2}{p_1} \right]$$

We know that,

$$Pv = mRT$$

So

$$W = mRT \ln \left[\frac{p_2}{p_1} \right]$$

b) **Work done during polytropic compression ($pv^n = \text{constant}$);**

The p-v diagram for a single stage single acting reciprocating air compressor is shown in fig.

The sequences of operations as represented on the diagram are as follows:

Process 4-1 => Represents the suction of air at pressure p_1

Process 1-2 => Air is compressed Polytropically from pressure p_1 to pressure p_2

Process 2-3 => Represents the discharge of air at pressure p_2

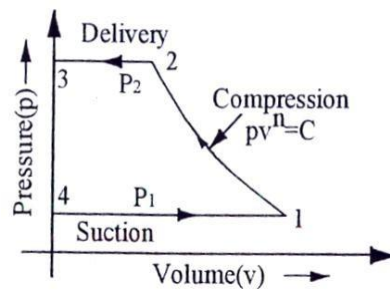


Fig. Polytropic compression

Work done = Area 1 – 2 – 3 – 4 – 1

$$W = W_{comp} + W_{Delivery} - W_{Suction} \quad \left[\because \text{For Polytropic process} \right]$$

$$W_{comp} = \frac{p_2 v_2 - p_1 v_1}{n-1}$$

$$\begin{aligned} W &= \frac{p_2 v_2 - p_1 v_1}{n-1} + p_2 v_2 - p_1 v_1 \\ &= \frac{p_2 v_2 - p_1 v_1 + (n-1)(p_2 v_2) - (n-1)(p_1 v_1)}{n-1} \\ &= \frac{p_2 v_2 - p_1 v_1 + n p_2 v_2 - p_2 v_2 - n p_1 v_1 + p_1 v_1}{n-1} \\ &= \frac{n p_2 v_2 - n p_1 v_1}{n-1} = \frac{n (p_2 v_2 - p_1 v_1)}{n-1} \end{aligned}$$

$$W = \frac{n}{n-1} [p_2 v_2 - p_1 v_1]$$

$$p_1 v_1 = mRT_1; \quad p_2 v_2 = mRT_2$$

$$W = \frac{n}{n-1} [mRT_2 - mRT_1]$$

$$W = mR T_1 \left[\frac{T_2}{T_1} - 1 \right]$$

For polytropic process,

$$\frac{T_2}{T_1} = \left(\frac{p_2}{p_1} \right)^{\frac{n-1}{n}}$$

$$W = \frac{n}{n-1} mR T_1 \left[\left(\frac{p_2}{p_1} \right)^{\frac{n-1}{n}} - 1 \right]$$

$$W = \frac{n}{n-1} p_1 v_1 \left[\left(\frac{p_2}{p_1} \right)^{\frac{n-1}{n}} - 1 \right] \quad [\because pv = mRT]$$

c) **Work done during isentropic compression (or) adiabatic compression [$pv^\gamma = C$]:**

The p-v diagram for a single stage single acting reciprocating air compressor is shown in fig.

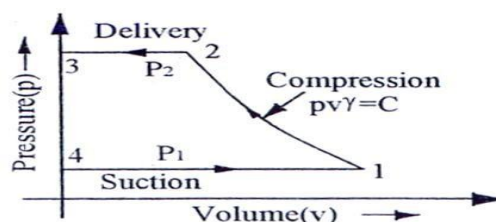


Fig. Adiabatic compression

The sequences of operation as represented on the diagram are as

Follows:

Process 4-1 => Represents the suction of air at pressure p_1

Process 1-2 => Air is compressed Polytropically from pressure p_1 to pressure p_2

Process 2-3 => Represents the discharge of air at pressure p_2

Work done = Area 1-2-3-4-1

$$W = W_{comp} + W_{Delivery} - W_{Suction} \quad \left[\because \text{For isentropic process} \right]$$

$$W_{comp} = \frac{p_2 v_2 - p_1 v_1}{\gamma - 1}$$

$$\begin{aligned} &= \frac{p_2 v_2 - p_1 v_1}{\gamma - 1} + p_2 v_2 - p_1 v_1 \\ &= \frac{p_2 v_2 - p_1 v_1 + (\gamma - 1)(p_2 v_2) - (\gamma - 1)(p_1 v_1)}{\gamma - 1} \\ &= \frac{p_2 v_2 - p_1 v_1 + \gamma p_2 v_2 - p_2 v_2 - \gamma p_1 v_1 + p_1 v_1}{\gamma - 1} \\ &= \frac{\gamma(p_2 v_2 - p_1 v_1)}{\gamma - 1} \end{aligned}$$

$$W = \frac{\gamma}{\gamma - 1} [p_2 v_2 - p_1 v_1]$$

$$p_1 v_1 = mRT_1; \quad p_2 v_2 = mRT_2$$

$$W = \frac{\gamma}{\gamma - 1} [mRT_2 - mRT_1]$$

$$W = \frac{\gamma}{\gamma - 1} mR T_1 \left[\frac{T_2}{T_1} - 1 \right]$$

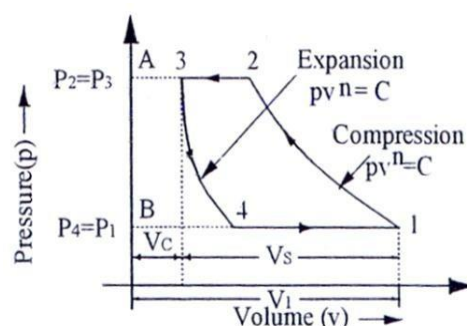
For polytropic process,

$$\frac{T_2}{T_1} = \left(\frac{p_2}{p_1} \right)^{\frac{\gamma - 1}{\gamma}}$$

$$W = \frac{\gamma}{\gamma - 1} p_1 v_1 \left[\left(\frac{p_2}{p_1} \right)^{\frac{\gamma - 1}{\gamma}} - 1 \right] \quad [\because pv = mRT]$$

Efficiency and work done by single stage reciprocating air compressor with clearance volume

When the piston reaches top dead centre in the cylinder there is a dead space between piston top and the cylinder head. This space is known as clearance space and the volume occupied by this space is known as clearance volume (v_c). Consider a reciprocating air compressor with clearance volume (v_c) as shown in fig



$P_1, v_1, T_1 \Rightarrow$ Initial pressure, Initial temperature, initial volume of air respectively

$P_2, v_2, T_2 \Rightarrow$ Corresponding values for the final conditions

$V_c \Rightarrow$ clearance volume

$V_s \Rightarrow$ Stroke volume = $V_1 - V_c$

$n \Rightarrow$ Polytropic index for compression and expansion

Work done by the compressor per cycle

$W = \text{Area } 1-2-3-4-1 = (\text{Area } 1-2-A-B-1) - (\text{Area } 3-A-B-4-3)$

$W =$ Work done during compression – work done during expansion

$$W = \frac{n}{n-1} p_1 v_1 \left[\left(\frac{p_2}{p_1} \right)^{\frac{n-1}{n}} - 1 \right] \quad W = \frac{n}{n-1} p_1 v_4 \left[\left(\frac{p_3}{p_4} \right)^{\frac{n-1}{n}} - 1 \right]$$

we know that,

$$p_1 v_1^n = p_2 v_2^n; \quad p_1 v_1^n = p_3 v_3^n$$

$$W = \frac{n}{n-1} p_1 v_1 \left[\left(\frac{p_2}{p_1} \right)^{\frac{n-1}{n}} - 1 \right] \quad W = \frac{n}{n-1} p_1 v_4 \left[\left(\frac{p_3}{p_4} \right)^{\frac{n-1}{n}} - 1 \right]$$

$$W = \frac{n}{n-1} p_1 \left[\left(\frac{p_2}{p_1} \right)^{\frac{n-1}{n}} - 1 \right] [v_1 - v_4]$$

$$W = \frac{n}{n-1} p_1 [v_1 - v_4] \left[\left(\frac{p_2}{p_1} \right)^{\frac{n-1}{n}} - 1 \right]$$

$$W = \frac{n}{n-1} p_1 v_a \left[\left(\frac{p_2}{p_1} \right)^{\frac{n-1}{n}} - 1 \right]$$

Where, $V_a = V_1 - V_4$ is the actual volume of free air delivered per cycle

$$W = \frac{n}{n-1} m R T_1 \left[\left(\frac{p_2}{p_1} \right)^{\frac{n-1}{n}} - 1 \right] \quad [P_1 V_a = m R T_1]$$

VOLUMETRIC EFFICIENCY

Volumetric efficiency is defined as the ratio of volume of free air sucked into the compressor per cycle to the stroke volume of the cylinder.

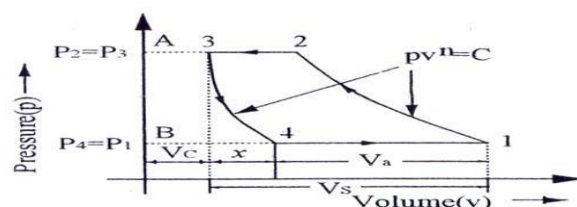
$$\eta_{\text{vol}} = \frac{\text{Volume of Free taken per cycle}}{\text{Stroke volume of the cylinder}}$$

$$\eta_{\text{vol}} = \frac{v_a}{v_s}$$

From $P - V$ diagram

$$V_a = v_s - x,$$

$$x = v_4 - v_c$$



volumetric efficiency

$$\eta_{vol} = \frac{v_s - v_c}{v_s} = \frac{v_s - (v_4 - v_c)}{v_s} = \frac{v_s - v_c \left[\frac{v_4}{v_c} - 1 \right]}{v_s}$$

$$\eta_{vol} = 1 - \frac{v_c}{v_s} \left[\frac{v_4}{v_c} - 1 \right]$$

Compression and expansion follows $PV^n = C$

$$P_3 V_3^n = P_4 V_4^n$$

$$\frac{v_4}{v_3} = \left(\frac{P_3}{P_4} \right)^{\frac{1}{n}}$$

From p-v diagram we know that,

$$V_3 = v_c, P_4 = P_1, P_3 = P_2$$

$$\frac{v_4}{v_c} = \left(\frac{P_2}{P_1} \right)^{1/n} \Rightarrow \frac{v_4}{v_c} \left(\frac{P_2}{P_1} \right)^{1/n}$$

Apply $\frac{v_4}{v_c}$ value in Equation (10)

$$\eta_{vol} = 1 - \frac{v_c}{v_s} \left[\left(\frac{P_2}{P_1} \right)^{\frac{1}{n}} - 1 \right]$$

Clearance ratio is defined as the ratio of clearance volume to swept volume.

$$\text{Clearance ratio, } C = \frac{v_c}{v_s}$$

$$\eta_{vol} = 1 - C \left[\left(\frac{P_2}{P_1} \right)^{\frac{1}{n}} - 1 \right]$$

Centrifugal compressors

In this type of compressor air enters axially and leaves radially.

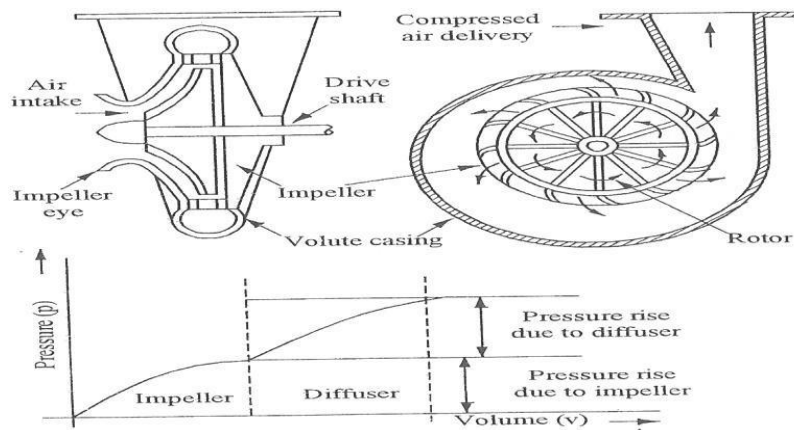
Construction:

- ❖ The arrangement of centrifugal compressor is shown in Fig.
- ❖ It consists of a rotating impeller, a casing and a diffuser.
- ❖ The impeller consists of a disc on which radial blades are attached. The impeller is surrounded by casing.
- ❖ The diffuser is other important part of the compressor, which is used to convert kinetic energy of air into pressure energy.
- ❖ The air coming out from the diffuser is collected in the casing and taken out from the outlet of the compressor.
- ❖ The impeller of a centrifugal compressor can be run at speeds of 20,000 to 30,000rpm

Working:

- ❖ When the power is given to compressor, the impeller rotates, and it sucks the air.
- ❖ This air enters axially with low velocity. The velocity and pressure of the air passing through the impeller are partially increased.

- ❖ Then the air is entered into diffuser. In the diffuser, kinetic energy is converted into pressure energy. So the pressure of air is further increased.
- ❖ Finally the air at high pressure is delivered to the receiver. Nearly half of the total pressure rise is achieved in the impeller and remaining half in the diffuser.
- ❖ The change of pressure and velocity of air passing through the impeller and diffuser are shown in Fig



Applications:

- ❖ Centrifugal compressors are suitable for super charging I.C. engines, refrigeration and low-pressure units

Vane blower

Construction:

- ❖ The arrangement of vane blower is shown in Fig.
- ❖ It consists of rotating drum, spring-loaded vanes, inlet and outlet ports and casing.

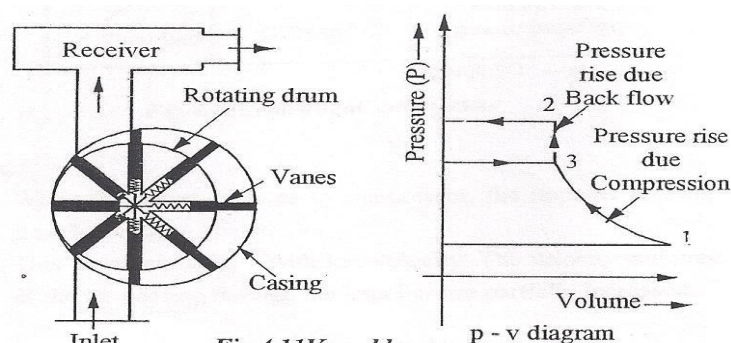


Fig. Vane blower

- ❖ The rotor is located eccentrically inside the casing.
- ❖ The rotor carries a set of spring-loaded vanes.
- ❖ These spring-loaded vanes are made of fiber of carbon.

Working:

- ❖ When the power is given to a vane blower, rotating drum (rotor) rotates, and the air is trapped between two consecutive vanes.
- ❖ As the rotation takes place the trapped air first expands and then gradually compressed due to decreasing volume between the rotor and outer casing.
- ❖ This partially compressed air is delivered to the receive.
- ❖ When the outlet is opened, there is a back flow of high-pressure air from receiver will rush back and mixed up with the entrapped air. So partially compressed air pressure is further increased. Finally high-pressure air is delivered from the receiver.
- ❖ In vane blower the pressure of air is increased first by decreasing the volume and then by back flow of air as shown in p-v diagram Fig

c) Roots blower:

Construction:

- ❖ The arrangement of Roots blower is shown in fig. which is simply a development of the gear pump.
- ❖ It consists of two rotors, which are aligned in different parallel axis.
- ❖ One of the rotors is connected to the drive and the second rotor is driven from the first.
- ❖ The two rotors rotate in opposite directions i.e. one in clockwise direction and the other in anti clockwise direction.
- ❖ The lobes of the rotors are of epicycloids, hypocycloid, or involutes profile to ensure correct matching.
- ❖ There must be small clearance between the lobe and casing to reduce the wear of moving parts.

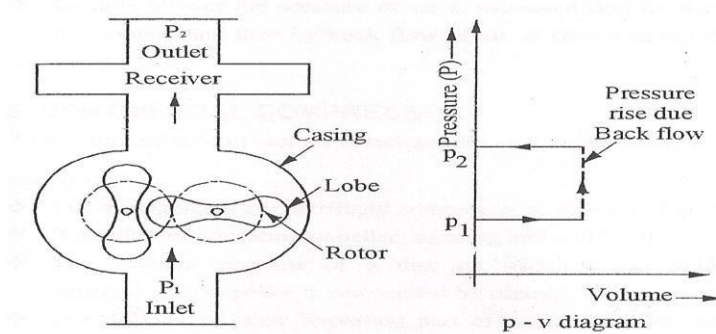


Fig. Roots blower

- ❖ When the power is given to the roots blower, rotors rotate and the air at atmospheric pressure is trapped between the lobe and casing.
- ❖ The trapped air moves along the casing and discharged into the receiver.
- ❖ The flow area from entry to exit remains constant. So, there is no developing in pressure.
- ❖ When the exit port opens, some high-pressure air from receiver will rush back and mixed up with entrapped air until the pressure is equalized.
- ❖ Thus the pressure of the entrapped air is increased by back flow of air.
- ❖ The $p-v$ diagram for this type of compressor is shown in Fig.

Differentiate between Reciprocating and Rotary compressor.

SL.NO	Rotary Compressor	Reciprocating Compressor
1.	Simple in construction	Complicated construction
2.	Speed is high	Speed is low
3.	It is suitable for large rates of flow at low discharge pressure	It is suitable for low rates of flow at very discharge pressure
4.	Maintenance cost is less	Maintenance cost is high
5.	There is no balancing problem	Balancing is major problem
6.	Simple lubrication system	More complicated
7.	Small in size for the same discharge compared with reciprocating compressors	Large in size for the same discharge comp with rotary compressor
8.	Uniform delivery of air	Delivery is not uniform

UNIT - 4

1. A single stage, single acting reciprocating air compressor has a bore of 200mm and a stroke of 300mm. It receives air at 1 bar and 20° C and delivers it at 5.5 bar. If the compression follows the law $p v^{1.3} = C$ and clearance volume is 5 % of the stroke volume. Determine: 1. the mean effective pressure and 2. Power required to drive the compressor if it runs at 500 r.p.m.

Given:

$$\begin{aligned} D &= 0.2\text{m} \\ L &= 0.3\text{m} \\ P_1 &= 1 \text{ bar} = 1 \times 10^5 \text{ N/m}^2 \\ T_1 &= 293 \text{ K} \\ P_2 &= 5.5 \text{ bar} = 5.5 \times 10^5 \text{ N/m}^2 \\ n &= 1.3 \\ V_C &= 5\% V_s \\ N &= 500 \text{ r.p.m} \end{aligned}$$

To Find:

$$\begin{aligned} P_m &=? \\ P &=? \end{aligned}$$

Solution

Stroke Volume

$$V_s = \frac{\pi}{4} \times D^2 \times L = 0.00942 \text{ m}^3$$

Clearance volume

$$V_C = 5\% V_s = 0.00047 \text{ m}^3$$

Initial Volume of air,

$$V_1 = V_C + V_s = 0.00989 \text{ m}^3$$

Expanded clearance Volume,

$$V_4 = V_C (P_2/P_1)^{1/n} = 0.00174 \text{ m}^3$$

Work Done

$$\begin{aligned} W &= \frac{n}{n-1} p_1 v_1 \left[\left(\frac{P_2}{P_1} \right)^{\frac{n-1}{n}} - 1 \right] \\ &= 1702 \text{ Nm} \end{aligned}$$

a. Mean Effective Pressure

$$P_m = \frac{\text{work done}}{\text{stroke volume}} = 1.807 \text{ bar}$$

b. Power required to drive the compressor

$$P = \frac{W \times N}{60} = 14.183 \text{ kW}$$

2. A two stage single acting reciprocating air compressor draws in air at a pressure of 1 bar and 17° C and compresses it to a pressure of 60 bar. After compression in the L.P cylinder, the air is cooled at a constant pressure of 8 bar to a temperature of 37° C. The L.P cylinder has a diameter of 150 mm and both the cylinder have 200mm stroke. If the law of compression is $p v^{1.35} = C$, find the power of the compressor, when it runs at 200 r.p.m. Take $R = 287 \text{ J/kg K}$.

Given:

$$\begin{aligned} P_1 &= 1 \times 10^5 \text{ N/m}^2 \\ P_2 &= 8 \times 10^5 \text{ N/m}^2 \\ P_3 &= 60 \times 10^5 \text{ N/m}^2 \\ T_1 &= 290 \text{ K} \\ T_3 &= 310 \text{ K} \\ D &= 0.15 \text{ m} \\ L &= 0.2 \text{ m} \\ n &= 1.35 \\ N &= 200 \text{ r.p.m} \end{aligned}$$

To Find:

$$\text{Power} = ?$$

Solution:

Volume of L.P cylinder

$$V_1 = \frac{\pi}{4} \times D^2 \times L = 0.0035 \text{ m}^3$$

Volume of H.P cylinder

$$\begin{aligned} \frac{P_1 V_1}{T_1} &= \frac{P_2 V_2}{T_3} \\ V_2 &= 0.00047 \text{ m}^3 \end{aligned}$$

Work done by L.P Cylinder

$$W_{L.P} = \frac{n}{n-1} p_1 v_1 \left[\left(\frac{P_2}{P_1} \right)^{\frac{n-1}{n}} - 1 \right]$$

$$W_{L.P} = 965 \text{ Nm}$$

Work Done By H.P Cylinder

$$W_{H.P} = \frac{n}{n-1} p_2 v_2 \left[\left(\frac{P_3}{P_2} \right)^{\frac{n-1}{n}} - 1 \right]$$

$$W_{H.P} = 996 \text{ Nm}$$

Total Work done by the Compressor

$$W = W_{L.P} + W_{H.P}$$

$$W = 1961 \text{ Nm}$$

a. Power

$$P = \frac{W \times N}{60} = 6.54 \text{ kW}$$

3. A centrifugal compressor runs at 15000 r.p.m and produce a stagnation pressure ratio of 4 between the impeller inlet and outlet. The stagnation conditions of air at the compressor intake are 1.01325 bar and 25° C respectively. The absolute velocity at the compressor inlet is axial. If the compressor has radial blades at the exit such that $V_{f2} = 135$ m/s and the total to total efficiency of the compressor is 0.78, draw the velocity triangle at exit of the rotor and compute the slip as well as the slip co-efficient. Rotor diameter at outlet is 580mm. Compute the absolute velocity at the compressor exit also.

Given:

$$N = 15000 \text{ r.p.m}$$

$$P_{02} / P_{01} = 4$$

$$P_{01} = 1.01325 \text{ bar}$$

$$T_{01} = 298 \text{ K}$$

$$\eta_{t-t} = 0.78$$

$$D_2 = 0.58 \text{ m}$$

$$V_{f2} = 135 \text{ m/s}$$

To find:

$$\text{Slip} = ?$$

$$\text{Slip co-efficient} = ?$$

$$V_2 = ?$$

Solution:

$$U_2 = \frac{\pi D_2 N}{60} = 455.53 \text{ m/s}$$

$$\frac{T_{02}'}{T_{01}} = \left\{ \frac{P_{02}}{P_{01}} \right\}^{\frac{\gamma-1}{\gamma}}$$

$$T_{02}' = 442.826 \text{ K}$$

$$\eta_{t-t} = \frac{T_{02}' - T_{01}}{T_{02} - T_{01}} = 0.78$$

$$T_{02} = 483.647 \text{ K}$$

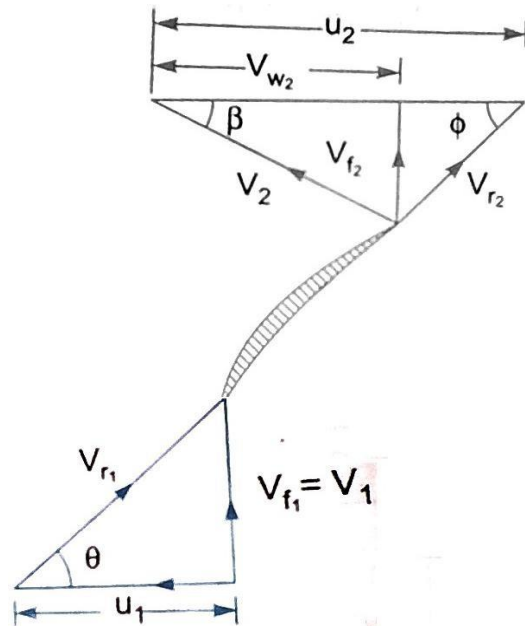
Actual work

$$W_{act} = C_p (T_{02} - T_{01}) = 186.603 \text{ KJ/kg}$$

We know that

$$W_{act} = V_{w2} U_2$$

$$V_{w2} = 409.6394 \text{ m/s}$$



a. Slip:

$$\text{Slip} = U_2 - V_{w2} = 45.89 \text{ m/s}$$

b. Slip co-efficient:

$$\text{Slip co-efficient} = V_{w2} / U_2 = 0.89925$$

c. Absolute velocity at exit:

$$V_2 = \sqrt{(V_{f2}^2 + U_2^2)} = 431.3113 \text{ m/s}$$

4. A centrifugal compressor compresses 30 kg of air per second. It runs at 15000 r.p.m. the air enters the compressor axially. The radius at exit of blade is 300 mm. The relative velocity of air at the tip is 100 m/s. The relative air angle at exit is 80°. Find the power and ideal head developed.

Given:

$$m = 30 \text{ kg/s}$$

$$N = 15000 \text{ r.p.m}$$

$$D_2 = 0.6 \text{ m}$$

$$V_{r2} = 100 \text{ m/s}$$

$$\phi = 80^\circ$$

To find

$$\text{Power developed} = ?$$

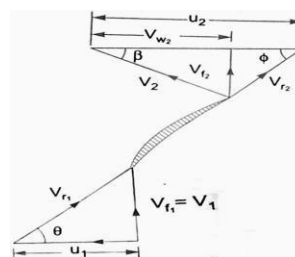
$$\text{Ideal Head} = ?$$

Solution

$$U_2 = \frac{\pi D_2 N}{60} = 471.2388 \text{ m/s}$$

$$\cos \phi = \frac{U_2 - V_{w2}}{V_{r2}}$$

$$V_{w2} = 453.8739 \text{ m/s}$$



a. Power developed

$$P = m \times V_{w2} \times U_2 = 6416.491 \text{ kW}$$

b. Ideal head developed

$$= V_{w2} \times U_2 = 213.883 \text{ KJ/kg}$$

5. The inlet stagnation conditions of a centrifugal compressor are 102 Kpa, 335 K. The hub and tip diameters at the impeller inlet are 100 mm, 250 mm respectively. The compressor runs at 120r.p.s and delivers 5 kg/s of air. Determine the air angle at inlet of the inducer blade and inlet mach number-relative velocity based.

Given:

$$\begin{aligned} P_{01} &= 102 \text{ Kpa} = 102 \text{ KN/m}^2 \\ T_{01} &= 335 \text{ K} \\ D_H &= 0.1 \text{ m} \\ D_T &= 0.25 \text{ m} \\ N &= 7200 \text{ r.p.m} \\ m &= 5 \text{ kg/s} \end{aligned}$$

To find:

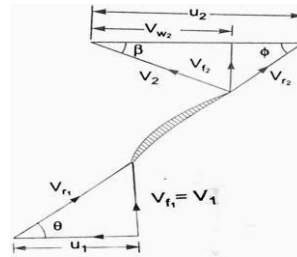
$$\begin{aligned} \theta &=? \\ M_1 &=? \end{aligned}$$

Solution

$$D_m = \frac{D_H + D_T}{2} = 0.175 \text{ m}$$

$$B = \frac{D_H - D_T}{2} = 0.075 \text{ m}$$

$$\rho_1 = \frac{P_{01}}{R \times T_{01}} = 1.060897 \text{ kg/m}^3$$



Mass flow rate

$$\begin{aligned} m &= \rho_1 [\pi D_m B] V_{f1} \\ V_{f1} &= \frac{114.3 \text{ m/s}}{\pi D_m N} \\ U_1 &= \frac{65.9734 \text{ m/s}}{60} \end{aligned}$$

a. Air angle at inlet

$$\tan \theta = \frac{V_{f1}}{U_1}, \theta = 60^\circ$$

b. Inlet mach number relative velocity based

$$\begin{aligned} \cos \theta &= U_1 / V_{r1} \\ V_{r1} &= 132 \text{ m/s} \\ M_1 &= \frac{V_{r1}}{\sqrt{\gamma R T_{01}}} = 0.366 \end{aligned}$$

6. A centrifugal blower runs at a speed of 3000 r.p.m. Its impeller outer diameter being 75 cm. The impeller blades are designed for a constant radial velocity of 57 m/s from the inlet to the outlet. There are no guide vanes so that the absolute velocity at inlet is axial. If the degree of reaction is 0.58 draw velocity triangle at the outlet and compute the exit blade angle. Find also the power input to the machine assuming a total –to–total efficiency of 0.75 and the exit pressure at the inlet to be 1 atm, and the total temperature to be 25°C.

Given:

$$\begin{aligned} N &= 3000 \text{ r.p.m} \\ D_2 &= 0.75 \text{ m} \\ V_{f1} = V_{f2} &= 57 \text{ m/s} \\ R_d &= 0.58 \\ \eta_{t-t} &= 0.75 \\ P_{01} &= 101 \text{ Kpa} = 101 \text{ KN/m}^2 \\ T_{01} &= 298 \text{ K} \end{aligned}$$

To find:

$$\begin{aligned} \phi &=? \\ P &=? \\ P_{02} &=? \end{aligned}$$

Solution

$$U_2 = \frac{\pi D_2 N}{60} = 117.8097 \text{ m/s}$$

Degree of reaction

$$R_d = 1 - \frac{V_{w2}}{U_2}$$

$$V_{w2} = 98.96 \text{ m/s}$$

a. From Velocity Triangle

$$\tan \phi = \frac{V_{f2}}{U_2 - V_{w2}}$$

$$\phi = 71.7^\circ$$

b. Power input to the blower

$$\begin{aligned} W_{act} &= V_{w2} U_2 = 11.658 \text{ KJ/kg} \\ \text{Power input to the blower} &= 11.658 \text{ KW/ Kg/s} \end{aligned}$$

c. The exit total pressure

$$\begin{aligned} W_{act} &= C_p (T_{02} - T_{01}) \\ (T_{02} - T_{01}) &= 11.6 \text{ K} \\ \eta_{t-t} &= \frac{T_{02}' - T_{01}}{T_{02} - T_{01}} = 0.75 \end{aligned}$$

$$(T_{02}' - T_{01}) = 8.7003$$

$$T_{02}' = 306.7 \text{ K}$$

$$\frac{T_{02}'}{T_{01}} = \left\{ \frac{P_{02}}{P_{01}} \right\}^{\frac{\gamma-1}{\gamma}}$$

$$P_{01} = 112.0623 \text{ KN/m}^2$$

7. Air flows through a blower; its total pressure is increased by 15 cm of water head. The inlet total pressure and temperature are 105 Kpa and 15°C. The total – to – total efficiency is 75 %. Evaluate (i) exit total pressure and temperature (ii) isentropic and actual change in total head enthalpy. Assume 50 % reaction.

Given

Height increased = 0.15 m

$P_{01} = 105 \text{ KN/m}^2$

$T_{01} = 288 \text{ K}$

$\eta_{t-t} = 0.75$

To find:

$P_{02} = ?$

$T_{02} = ?$

Isentropic change in enthalpy =?

Actual change in enthalpy =?

Solution

$$H = \frac{P}{\rho g}$$

Pressure increased = 1471 N/m²

$P_{02} = P_{01} + \text{pressure increased}$

$$P_{02} = 106.47 \text{ KN/m}^2$$

$$\frac{\gamma-1}{\gamma}$$

$$\frac{T_{02}'}{T_{01}} = \left\{ \frac{P_{02}}{P_{01}} \right\}$$

$$T_{02}' = 289.14 \text{ K}$$

$$\eta_{t-t} = \frac{T_{02}' - T_{01}}{T_{02} - T_{01}} = 0.75$$

$$T_{02} = 289.528 \text{ K}$$

Isentropic change in enthalpy

$$= C_p (T_{02}' - T_{01}) = 1.152 \text{ KJ/kg}$$

Actual Change in enthalpy

$$= C_p (T_{02} - T_{01}) = 1.535 \text{ KJ/kg}$$

8. A single acting reciprocating air compressor has cylinder diameter and stroke of 200 mm and 300 mm respectively. The compressor sucks air at 1 bar and 27° C and delivers at 8 bar while running at 100 r.p.m.

find: 1. Indicated power of the compressor 2. Mass of air delivered by the compressor per minute 3.

Temperature of air delivered by the compressor. The compression follows the law $p v^{1.25} = C$.

Given:

$D = 0.2 \text{ m}$

$L = 0.3 \text{ m}$

$P_1 = 1 \times 10^5 \text{ N/m}^2$

$T_1 = 300 \text{ K}$

$P_2 = 8 \times 10^5 \text{ N/m}^2$

$N = 100 \text{ r.p.m}$

$n = 1.25$

$R = 0.287 \text{ KJ/kg K}$

To find:

Indicated power =?

Mass delivered per minute =?

Temperature of air delivered =?

Solution:

$$V_1 = \frac{\pi \times D^2}{4} \times L = 0.0094 \text{ m}^3$$

a. Indicated power

$$W = \frac{n}{n-1} \times p_1 v_1 \left[\left(\frac{P_2}{P_1} \right)^{\frac{n-1}{n}} - 1 \right]$$

$$= 2425 \text{ Nm}$$

$$\text{Power} = \frac{W \times N}{60} = 4.042 \text{ kW}$$

b. Mass of air delivered per minute

$$P_1 V_1 = m R T_1$$

$$m = 0.0109 \text{ kg}$$

$$\text{Mass delivered per minute} = m \times N$$

$$= 1.09 \text{ kg}$$

c. Temperature of air delivered

$$(T_2/T_1) = (P_2/P_1)^{n-1/n}$$

$$T_2 = 454.8 \text{ K}$$